

Global CSR Hub

Azure-Integrated Architecture

Hackathon MVP — Technical Design Document

MVP Build	Azure Powered	Next.js 15	Convex Backend
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Executive Summary

This document describes the technical architecture for the **Global CSR Hub**, a hackathon MVP that connects corporations with vetted NGOs for transparent, AI-assisted CSR funding. The platform layers enterprise-grade **Microsoft Azure services** on top of a modern Next.js 15 + Convex real-time stack, delivering intelligent search, AI-powered matching, secure document management, and enterprise SSO in a single cohesive product.

4 Azure Services	5 AI Features	4 Blob Containers	7 Demo Steps
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System Architecture Overview

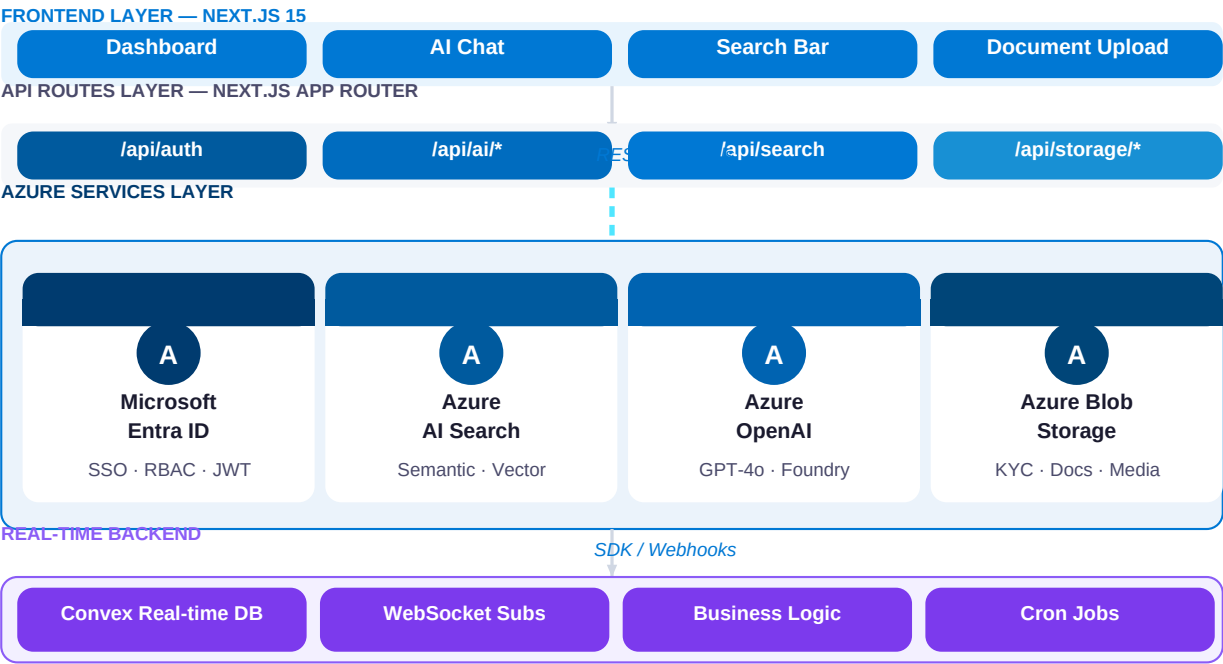


Figure 1 — Azure-integrated system architecture showing the four-layer stack: Next.js 15 frontend → API Routes → Azure Services → Convex real-time backend.

Technology Stack

Layer	Technology	Role
Frontend	Next.js 15 (App Router)	UI, routing, SSR
Real-time DB	Convex	Queries, mutations, WebSocket, cron
Auth (Public)	Clerk	NGO/public user sign-up
Auth (Enterprise)	Microsoft Entra ID	Corporate SSO, RBAC, JWT
AI Search	Azure AI Search	Full-text + semantic + vector search
AI / LLM	Azure OpenAI (GPT-4o)	Chatbot, summaries, matching, risk
Storage	Azure Blob Storage	KYC docs, proposals, reports, images
Payments	Stripe Connect	Funding transactions (placeholder)

Azure Service Integration Details

1 Azure Blob Storage — Document Management

Secure, scalable document storage using SAS (Shared Access Signature) tokens for direct client-to-cloud uploads, bypassing the application server and reducing latency.

Container	Purpose	Max Size	Access Policy
kyc-documents/	NGO KYC verification docs	50 MB/file	SAS token (1 hr)
proposal-files/	Proposal attachments & budgets	100 MB/file	SAS token (1 hr)
reports/	Audit & sustainability PDFs	200 MB/file	SAS token (1 hr)
profile-images/	Org logos & user avatars	5 MB/file	Public CDN

Upload Flow:

- ① Frontend requests a time-limited SAS token from `/api/storage/sas`
- ② Client uploads the file **directly** to Azure Blob Storage (server bypass)
- ③ The resulting Blob URL is persisted to the Convex document record
- ④ Downloads are served via SAS token with configurable expiry

2 Microsoft Entra ID — Enterprise Authentication

Dual-auth strategy: **Clerk** for fast NGO/public onboarding, **Microsoft Entra ID** for enterprise SSO. Entra ID group claims map directly to CSR Hub roles, enabling zero-friction corporate logins.

Step	Action	Implementation
1	Enterprise Login button clicked	Triggers MSAL redirect flow
2	Redirect to Entra ID	@azure/msal-browser consent screen
3	Authorization code returned	Callback to <code>/api/auth/entra/callback</code>
4	Code exchanged for tokens	@azure/msal-node server-side exchange
5	User synced to Convex	Upsert to users table with Entra ID
6	Role mapped from group claims	Entra groups → CORPORATION / ADMIN roles

3 Azure AI Search — Intelligent Search

Full-text, vector, and semantic search across all CSR data. Enables natural-language queries like *"water projects in Southeast Asia under \$500K"* with ranked, faceted, and highlighted results.

Index Name	Key Fields	Search Features
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ngo-index	name, type, country, sdgTags, trustScore, verificationStatus	Full-text · Filters · Facets
proposal-index	title, summary, category, sdgTags, budgetTotal, targetCountries, status	Semantic · Vector embeddings
country-index	name, code, region, subRegion, currency	Autocomplete · Suggestions
sdg-index	goal, title, description, targets	Semantic search

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Azure OpenAI / AI Foundry — AI Features

The most impactful demo feature. Five distinct AI capabilities powered by GPT-4o, including streaming chat, automated proposal summarization, and AI-driven matching.

Feature	Model	Purpose
CSR Chatbot	GPT-4o	Natural-language Q&A — platform, SDGs, NGOs
Proposal Summarizer	GPT-4o	Auto-generate executive summary on proposal creation
AI Matching Engine	GPT-4o + embeddings	Match corporations with top-fit proposals
Risk Assessment	GPT-4o	Analyze org profile → risk score + plain-English explanation
Report Generator	GPT-4o	Draft sustainability reports from structured data

Configuration & Dependencies

Environment Variables

Azure Blob Storage

AZURE_STORAGE_CONNECTION_STRING	DefaultEndpointsProtocol=https;AccountName=...
AZURE_STORAGE_ACCOUNT_NAME	csrhubstg
AZURE_STORAGE_ACCOUNT_KEY	

Microsoft Entra ID

AZURE_AD_CLIENT_ID	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
AZURE_AD_CLIENT_SECRET	
AZURE_AD_TENANT_ID	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
AZURE_AD_REDIRECT_URI	http://localhost:3000/api/auth/entra/callback

Azure AI Search

AZURE_SEARCH_ENDPOINT	https://csrhub-search.search.windows.net
AZURE_SEARCH_API_KEY	
AZURE_SEARCH_INDEX_NAME	ngo-index

Azure OpenAI

AZURE_OPENAI_ENDPOINT	https://csrhub-openai.openai.azure.com
AZURE_OPENAI_API_KEY	
AZURE_OPENAI_DEPLOYMENT_NAME	gpt-4o
AZURE_OPENAI_API_VERSION	2024-08-01-preview

NPM Dependencies

Package	Version	Purpose
@azure/storage-blob	^12.24	Blob Storage client — SAS tokens & upload
@azure/identity	^4.0	Azure credential management
@azure/search-documents	^12.0	AI Search SDK — indexing & queries
@azure/openai	^2.0	Azure OpenAI SDK — GPT-4o completions
@azure/msal-browser	^3.10	Client-side MSAL — Entra ID redirect flow
@azure/msal-node	^2.6	Server-side MSAL — token exchange
openai	^4.52	OpenAI SDK (streaming support)

Hackathon Demo Flow & Implementation Plan

Live Demo Script

1	Landing Page Browse NGOs & proposals powered by Azure AI Search with semantic ranking
2	Enterprise Login Click 'Enterprise SSO' → Microsoft Entra ID consent → corporate dashboard
3	Upload KYC Docs Drag-and-drop upload → SAS token generated → file lands directly in Azure Blob
4	AI Chatbot Ask: "Show me water projects in Indonesia under \$500K" → GPT-4o streaming response
5	AI Matching Corporation clicks 'Find Proposals' → top 5 recommendations with relevance scores
6	Proposal Summary Open any proposal → auto-generated executive summary via Azure OpenAI
7	Fund Project Stripe Connect placeholder → Convex trust score updates in real-time

Implementation Order (Hackathon Sprint)

Priority	Service	Reason
① High	Azure OpenAI	Most impressive demo feature — build chatbot + summary first
② High	Azure AI Search	Search bar with semantic ranking — second biggest WOW
③ Medium	Azure Blob Storage	Document upload with SAS tokens — practical & polished
④ Medium	Microsoft Entra ID	Enterprise SSO login — completes the corporate story
⑤ Low	Polish & Branding	Azure branding, demo script, screenshots & README

Note: This architecture document is prepared for hackathon evaluation purposes. All Azure service endpoints, keys, and connection strings shown are illustrative placeholders and must be replaced with real credentials before deployment.